

University Of California, Berkeley

Associate of Bioengineering | GPA: 4.0/4.0

Relevant coursework: MATLAB, 3D Modeling, Fluid Mechanics, Circuit Design, Bioinstrumentation analysis, Noise matching, Signal Acquisition, Data Acquisition system design.

Berkeley, California

Aug 2025 - Present

De Anza College

Associate of Bioengineering | GPA: 4.0/4.0

Honors: Student Athlete of the Year, Dean's List (2023-2025)

Relevant coursework: C++, Linear Algebra, Organic Chemistry, Physics for Engineers, Statistics, Circuit analysis

Cupertino, California

Sept 2023 - Jul 2025

SKILLS

Technical: Molecular cloning workflows, bacterial culture, mammalian cell culture, Gram/Ziehl–Neelsen staining, PCR/qPCR, Sanger/NGS sample preparation, Python, MATLAB, SolidWorks CAD, Microsoft Excel, scientific reporting.

WORK EXPERIENCE

Nam Khoa Biotek Company

Microbiologist Intern

- Executed 100+ clinical diagnostics, including Gram and Ziehl–Neelsen staining, TB screening, and Streptococcus testing for childbirth safety, followed by targeted antibiograms and microscopic validation.
- Performed high-throughput antibiotic susceptibility testing (MIC assays, *S. aureus* profiling) on ~30 samples daily while maintaining quality control and optimizing workflows, reducing turnaround time by 30% in a four-member team.

Ho Chi Minh City, Viet Nam

Jun 2024 - Aug 2024

Center of CRISPR Target Discovery

Research Associate

- Support pooled CRISPR–Cas9 functional genomic screening workflows, with emphasis on molecular cloning, gRNA library preparation, plasmid assembly, PCR amplification, and gel electrophoresis for construct validation.
- Conduct cloning-related workflows including bacterial transformation, colony growth/expansion, plasmid amplification, PCR screening, and Sanger sequencing analysis to verify successful constructs.
- Prepare CRISPR screen samples for downstream NGS workflows, including library amplification, cleanup, quantification, and sequencing submission preparation.
- Perform mammalian cell culture with Jurkat and HCT116 cells, including passaging, thawing/freezing, cryopreservation, genomic DNA extraction, and execution of CRISPR protocols using established SOPs.

Berkeley, California

Oct 2025 - Present

RESEARCH EXPERIENCE

Moriel Vandsburger Lab, UC Berkeley

Undergraduate Researcher

- Fabricate MRI phantoms across varying pH conditions to support cardiac metabolic imaging (creatine studies).
- Collaborate on AAV-based gene therapy experiments in murine models, assisting in evaluating in vivo delivery and expression.
- Perform qPCR on mouse liver samples to quantify DNA and assess transduction efficiency, alongside protein analysis (ELISA) and tissue processing to evaluate therapeutic outcomes

Berkeley, California

Jan 2026 - Present

PROJECT

CNT Integrated Impedance Microfluidic Sensor for ctDNA qualification

- Designed a multilayer microfluidic ctDNA sensing platform integrating screen-printed CNT electrodes, AC impedance circuitry, and purification/denaturation workflows for quantitative biomarker detection.
- Modeled signal optimization and sample processing, achieving ~85–90% processing efficiency, 2× signal-to-noise improvement, and 20–30% impedance differentiation across target concentrations.

Dec, 2025

Carbon Nano Tubes in Biosensor

- Designed a multi-channel CNT-based biosensor capable of detecting up to six molecular biomarkers linked to diabetes and cancer, integrating microfabrication and circuit analysis to model a functional prototype under review by Jadoo Technology.

Oct, 2025

BreathBiomics Gas-Sensor Array

- Engineered an ESP32-powered BreathBiomics gas-sensor array integrating MQ-series and Bosch sensors with sensor-fusion and signal-analysis workflows to profile disease-associated VOC signatures in real time as a portable proof of concept for noninvasive breath diagnostics.

Dec, 2025

Differential Pressure Spirometer with Analog Front End

- Designed and prototyped a spirometer using MPX2010DP differential pressure sensing, instrumentation amplification, and analog filtering, achieving respiratory waveform capture with ~350 Hz sampling and ~1000× front-end gain.
- Improved signal fidelity with SNR budgeting, oversampling, and 60 Hz noise suppression, achieving ~2× SNR improvement, <15% measurement variability, and stable flow signal detection across repeated breathing trials.

Apr, 2026